

1.1

2.

(a)

J : John is telling the truth.

B : Bill is telling the truth.

$$(J \wedge B) \vee (\neg J \wedge \neg B)$$

(b)

F : I'll have fish.

C : I'll have chicken.

M : I'll have mashed potatoes.

$$(F \vee C) \wedge \neg(F \wedge M)$$

(c)

A : 3 is a divisor of 6.

B : 3 is a divisor of 9.

C : 3 is a divisor of 15.

$$A \wedge B \wedge C$$

5.

(a) Well-formed.

(b) Not well-formed, as commas don't mean anything in sentential notation.

(c) Well-formed.

(d) Not well-formed, as there's no symbol between the two groups of parentheses.

8.

(a) Taxes will go up or the deficit will go up.

(b) $\neg(\neg T \wedge \neg D) = (T \vee D)$

Taxes and the deficit won't both go up, but either taxes or the deficit will go up.

- (c) Either taxes will increase and the deficit won't, or the deficit will increase and taxes won't.

9.

(a)

Premises:

- The main course will be either beef or fish.
- The vegetable will be either peas or corn.
- We will not have both fish as the main course and corn as the vegetable.

Conclusion:

- We will not have both beef as the main course and peas as the vegetable.

This conclusion is incorrect, because having both beef as the main course and peas as the vegetable still satisfies all of the premises.

(b)

Premises:

- Either John or Bill is telling the truth.
- Either Bill or Sam is lying.

Conclusion:

- Either John is telling the truth or Sam is lying.

This conclusion is correct:

- If John is lying, then Bill must be telling the truth. If Bill is telling the truth, then Bill isn't lying, which means Sam is.
- If Sam is telling the truth, then Bill must be lying. If Bill is lying, then he isn't telling the truth, which means John is.

1.2

2.

(a)

P	Q	$\neg P$	$(Q \vee \neg P)$	$P \wedge (Q \vee \neg P)$	$\neg(P \wedge (Q \vee \neg P))$
T	T	F	T	T	F
T	F	F	F	F	T
F	T	T	T	F	T
F	F	T	T	F	T

(b)

P	Q	R	$\neg P$	$(P \vee Q)$	$(\neg P \wedge R)$	$(P \vee Q) \wedge (\neg P \wedge R)$
T	T	T	F	T	T	T
T	T	F	F	T	F	F
T	F	T	F	T	T	T
T	F	F	F	T	F	F
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	T	F	T	F
F	F	F	T	F	T	F

3.

(a)

P	Q	$P + Q$
T	T	F
T	F	T
F	T	T
F	F	F

(b)

$$(P \vee Q) \wedge \neg(P \wedge Q)$$

P	Q	$P \vee Q$	$\neg(P \wedge Q)$	$(P \vee Q) \wedge \neg(P \wedge Q)$
T	T	T	F	F
T	F	T	T	T
F	T	T	T	T
F	F	F	T	F

4.

$$\neg(\neg P \wedge \neg Q)$$

P	Q	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$	$\neg(\neg P \wedge \neg Q)$	$P \vee Q$
T	T	F	F	F	T	T
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	T	F	F

6.

P	Q	$P Q$	$P \wedge Q$	$\neg(P \wedge Q)$
T	T	F	T	F
T	F	T	F	T
F	T	T	F	T
F	F	T	T	F

8.

A and C are equivalent.

B and E are equivalent.

(a)

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$(\neg P \wedge \neg Q)$	$(P \wedge Q) \vee (\neg P \wedge \neg Q)$
T	T	F	F	T	F	T
T	F	F	T	F	F	F
F	T	T	F	F	F	F
F	F	T	T	F	T	F

(b)

P	Q	$\neg P$	$\neg Q$	$\neg P \wedge Q$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

(c)

P	Q	$\neg P$	$\neg Q$	$P \wedge \neg Q$	$(\neg P \wedge Q)$	$(P \wedge \neg Q) \wedge (\neg P \wedge Q)$
T	T	F	F	T	T	T
T	F	F	T	T	F	F
F	T	T	F	F	T	F
F	F	T	T	T	T	T

(d)

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$\neg(P \wedge Q)$
T	T	F	F	T	F
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	F	T

(e)

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$\neg P \vee (P \wedge Q)$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	F	T
F	F	T	T	F	T